

Solution to Exercise Sheet: Application Layer

Exercise 1

1.1: What are sockets used for?

Sending and receiving data from the network. They serve as API communication endpoints that allow processes to send and receive data across networks.

1.2: What does HTTP stand for and what is the protocol used for?

HTTP stands for Hypertext Transfer Protocol. It's the Internet's primary application layer protocol for transferring web content.

1.3: Is the following statement true for HTTP?

"HTTP is stateless by default, but statefulness can be enabled by the client"

Give a brief explanation of your answer.

This statement is not correct. HTTP is an inherently stateless protocol; the server maintains no information about past client requests. However, session states can be kept in cookies which are communicated as part of the HTTP headers. This does not affect the protocol itself though which remains stateless even when cookies are used.

1.4: What is the difference between persistent and non-persistent HTTP?

In persistent HTTP, multiple objects can be loaded through the same TCP connection.

1.5: What is the difference between the two HTTP status codes 403 and 404?

- **403 Forbidden: The client is not authorised to access the resource, but it can be found.**
- **404 Not Found: The requested resource id not found on the server.**

1.6: Why was HTTP/2 introduced?

To mitigate HOL Blocking. HTTP/2 was introduced to decrease the delay in multi-object HTTP requests. It addresses the head-of-line blocking issues in HTTP/1 & 1.1.

1.7: What is the purpose of web caches?

Satisfying client requests without involving the origin server.

1.8: Is it true that SMTP is used to send, receive, and access emails? Give a brief explanation of your answer.

This is partially true. SMTP is only used for sending and transferring email messages between the sending UA and mail servers. For accessing and retrieving emails from servers, separate protocols, such as IMAP (Internet Mail Access Protocol) or web-based interfaces using HTTP, are used.

1.9: What is the difference between the approaches to trusting keys in OpenPGP and S/MIME?

S/MIME uses a public key infrastructure with CAs, and OpenPGP uses a web-of-trust approach

Exercise 2

2.1: Explain the term “Client-Server Paradigm”.

The Client-Server Paradigm in networking is where the client initiates requests for services/resources and the server listens for these requests, processes them and sends back the appropriate response.

2.2: Explain the term “Peer-to-peer Architecture.”

P2P architecture is where arbitrary end systems communicate directly to each other, peers act as both server and client, thus peers request service from other peers and provide service in return to other peers.

2.3: Explain the term “elastic” in the context of network throughput.

In the context of network throughput, refers to the independence of a network applications to differences in network bandwidth.

2.4: Explain the difference between iterated and recursive queries in DNS.

A recursive query puts full burden of the name resolution on a contacted name server. Each server in the DNS hierarchy will forward the request to the next level until the request reaches the authoritative DNS server. Then the response travels back up the hierarchy through all intermediary DNS servers back to the client. This process simplifies resolution for the client but increases workload for the DNS resolver. In iterated queries, on the other hand, the contacted DNS servers in the hierarchy will return as a response the next DNS server to contact, and contact has to be established by the client (or the local DNS server).

Exercise 3

In this exercise you will make first experiences with capturing and analysing network data.

3.0: Download & install Wireshark on your laptop/PC from: <https://www.wireshark.org/download.html>. Then, familiarize yourself with the Wireshark UI and how to capture live network data using chapters 3&4 of its documentation: https://www.wireshark.org/docs/wsug_html_chunked/index.html

3.1: Start a live network data capture. While running the live capture, visit the website <http://example.com> in your browser. Then wait a few moments and reload the page in your browser. Then stop the capture. By looking at the captured data, answer the following questions:

- a) How long did it take for the server to respond to the first HTTP GET with an HTTP OK?
0.106148 seconds

- b) What is the IP address of your computer?

IP address: 10.XX.XX.X07

- c) What is the IP address of the example.com server?

93.184.215.14

- d) Is your browser running HTTP version 1.0, 1.1, or 2? What version of HTTP is the server running?

HTTP/1.1

- e) What is the HTTP status code returned by the server in the first HTTP GET request?

200 OK

- f) How many bytes of content are being returned to your browser in the response to the first HTTP GET request?

648 bytes / 1256 bytes

- g) How long is the data that was sent to the browser as response to the first HTTP GET request valid if it is cached? Which header field tells you this info?

Cache-control: max-age=604800

- h) How does the HTTP GET request to the server change when reloading the page? How does the server's response change from the original response?

The first original response was 200 OK then the later changed to 304 Not Modified. The 304 does not include any data in the body.

- i) Find the first DNS query involving the domain example.com. To which IP address was the query sent?

10.220.2.5

- j) What type is specified in the DNS query?

Type A

- k) What is the answer data in the response to the query?

example.com: type A, class IN, addr 93.184.215.14

Note!! The answers above can be found in the attached packet tracers on itsLearning. The answers will differ based on your devices.